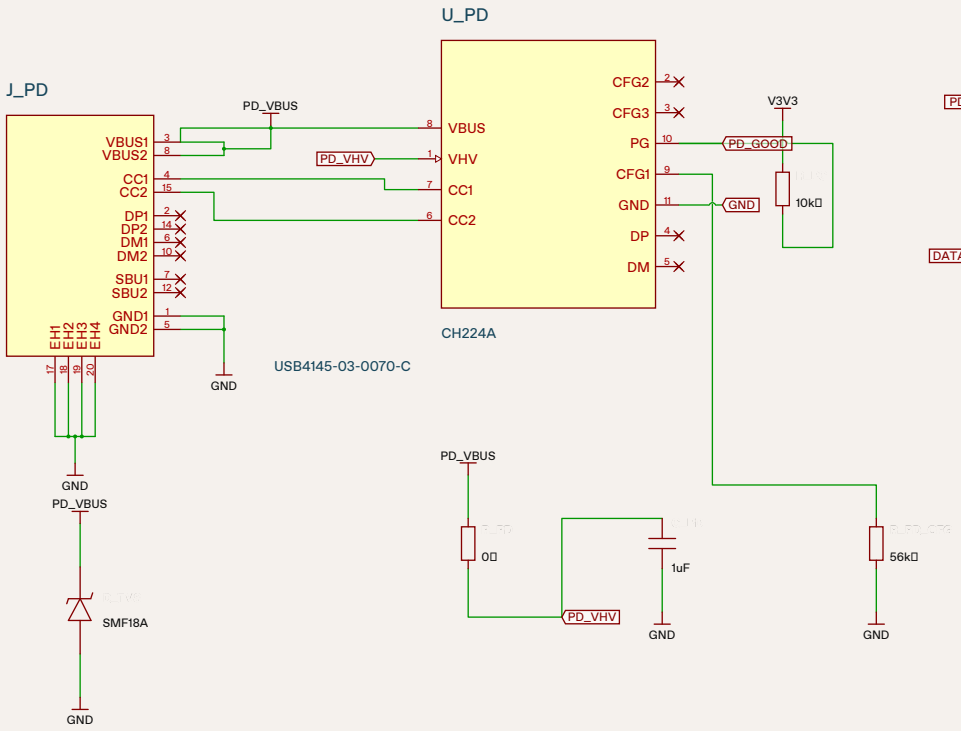
NEMA11 / RP2040

Power & USB-PD

Motor supply negotiation, reverse-blocked logic power and 3.3 V conversion

USB-C motor power

Fixed 15 V request; data and SBU contacts unused



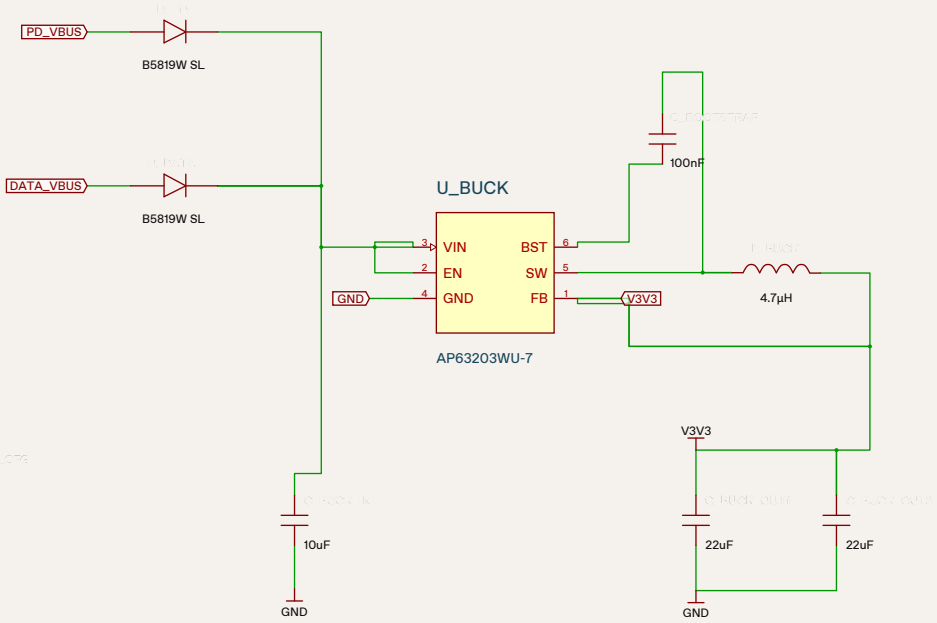
CH224A: 56 kΩ CFG1-to-GND selects 15 V; CFG2/3 NC.

C_PD: 1 µF / 50 V (C15849); R_PD: 0 Ω (C21189).

PD_VBUS = motor rail DATA_VBUS = programming USB 5 V

3.3 V logic supply

AP63203 fixed-output buck; USB power is diode-isolated

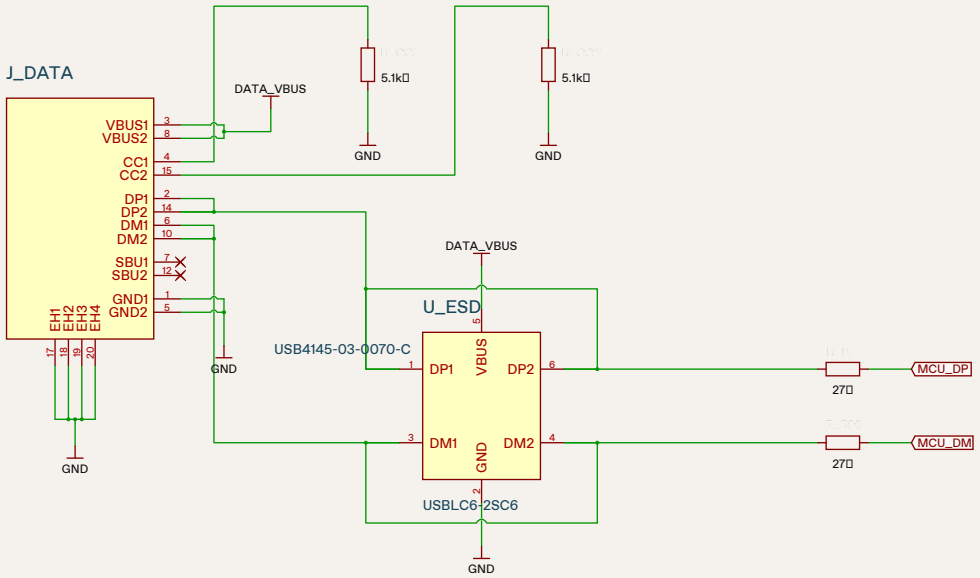


VIN range and transients require bench qualification.

Programming USB-C, data-line protection and nonvolatile boot interfaces

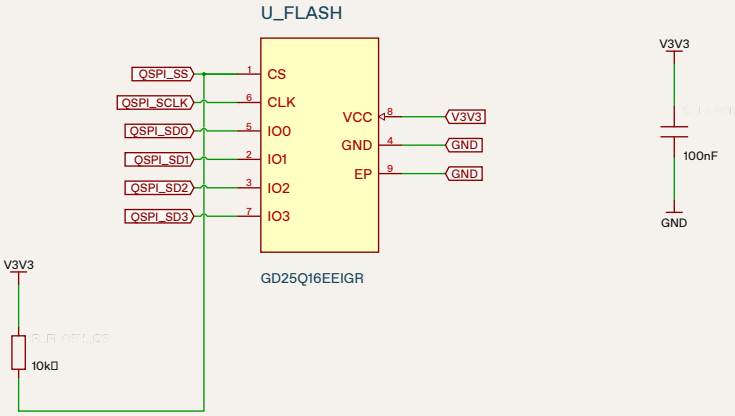
USB-C data port

Separate 5.1 kΩ CC pull-downs; duplicated D+/D− contacts



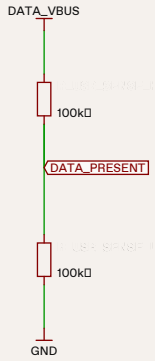
QSPI flash & boot selection

GD25Q16E; BOOT contact asserts the boot input



USB presence

Resistive divider to MCU; fixed 5 V input only



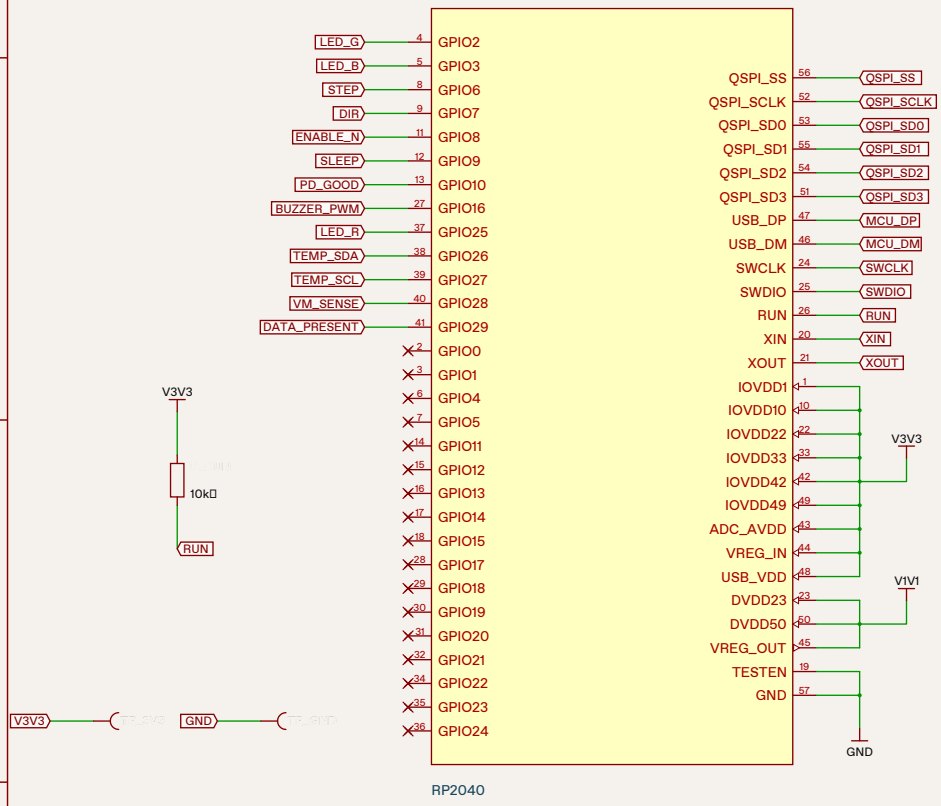


NEMA11 / RP2040

Controller & clocks

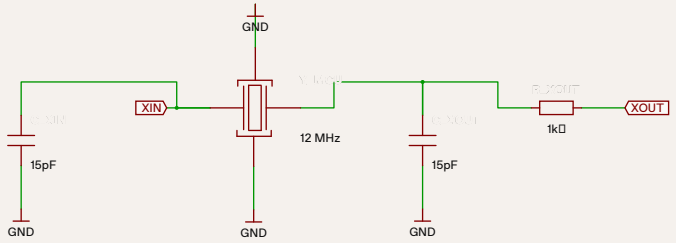
MCU interfaces, reset/debug access, power decoupling and clock support

U_MCU



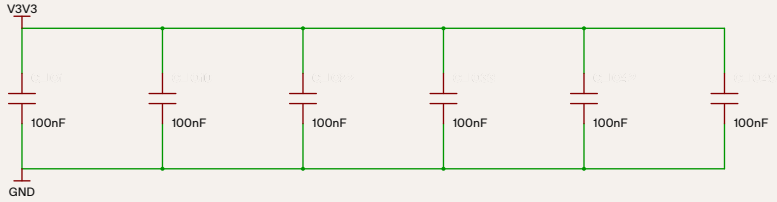
12 MHz crystal

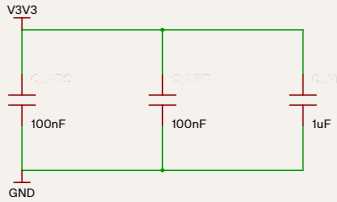
Native 4-pin crystal; load capacitors and series resistor



3.3 V bypass

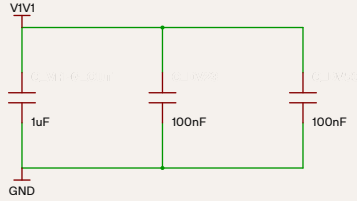
IOVDD, ADC and USB decoupling





1.1 V core

On-chip VREG_OUT feeds DVDD



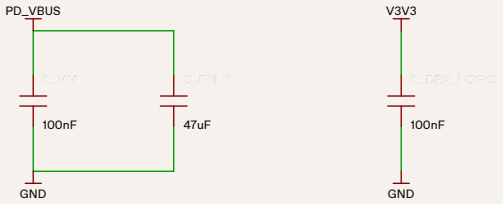
NEMA11 / RP2040

Stepper motor driver

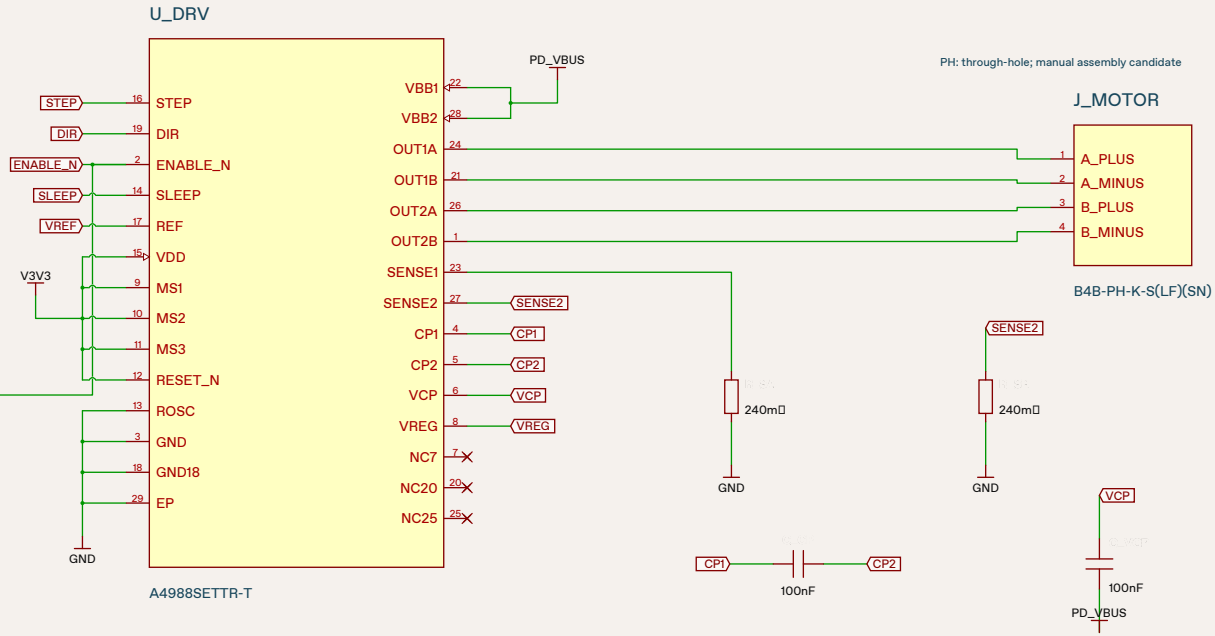
A4988 phase outputs, charge pump, current limit and motor-voltage measurement

Driver supplies

Local bulk and high-frequency bypass



JST-PH pin order: 1 A+ 2 A– 3 B+ 4 B–

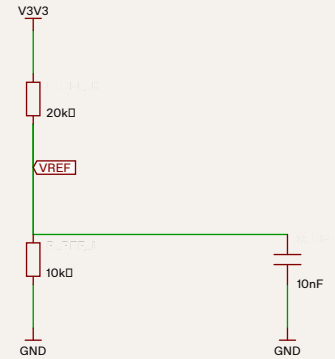


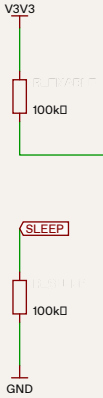
PH: through-hole; manual assembly candidate

B4B-PH-K-S(LF)(SN)

Current reference

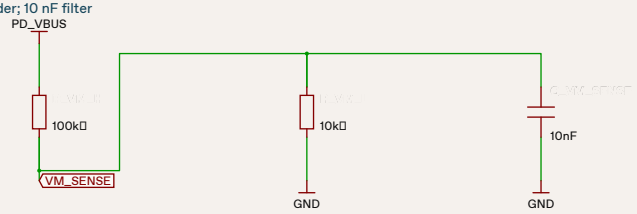
VREF / (8 × RSENSE) = 0.573 A nominal; verify waveform and thermal limits





Motor voltage monitor

100 kΩ / 10 kΩ divider; 10 nF filter





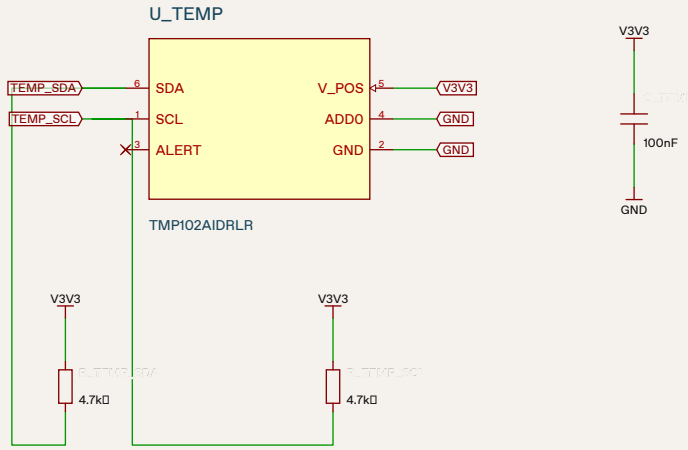
NEMA11 / RP2040

Temperature & feedback

I²C temperature telemetry, RGB status outputs and PWM magnetic buzzer

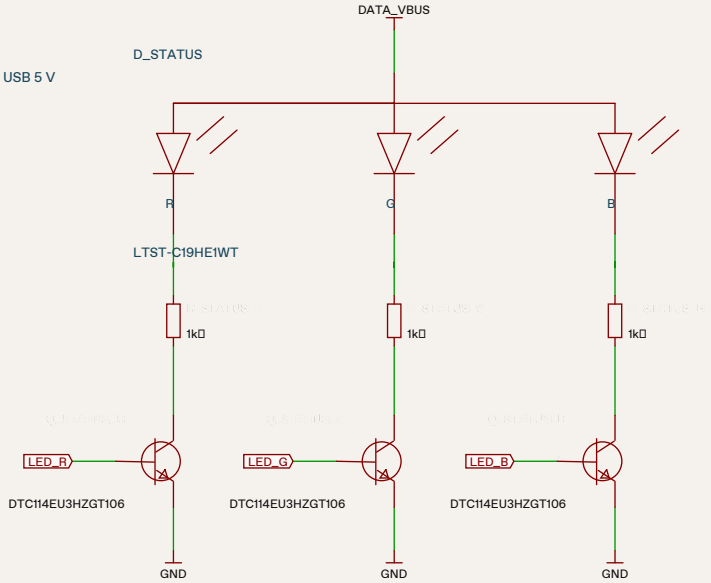
TMP102 temperature

ADD0 grounded: address 0x48; ALERT unused



RGB status indicator

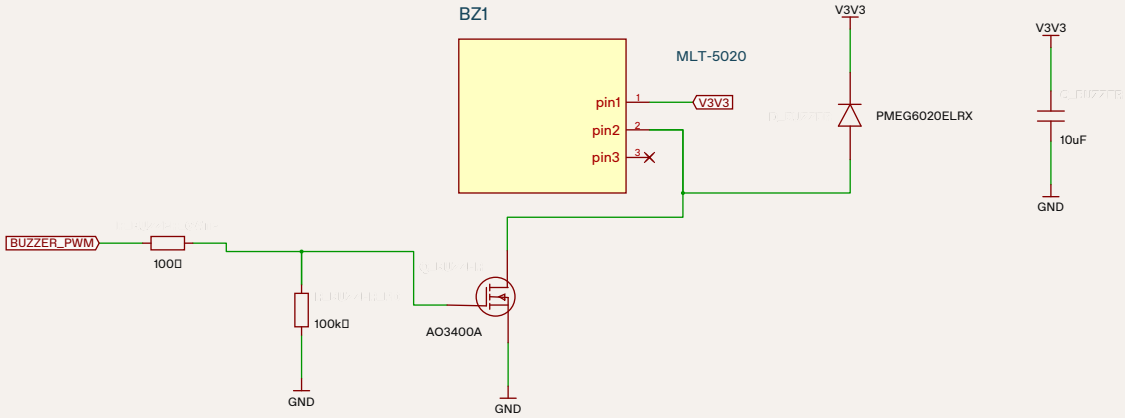
Common anode uses programming USB 5 V



DTC114E: NPN with internal input/bias resistors.

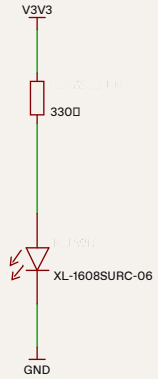
PWM buzzer

MLT-5020: 4 kHz, 50%; 100 mA max at 3 V test



Power indication

3.3 V rail present; RGB/buzzer require firmware

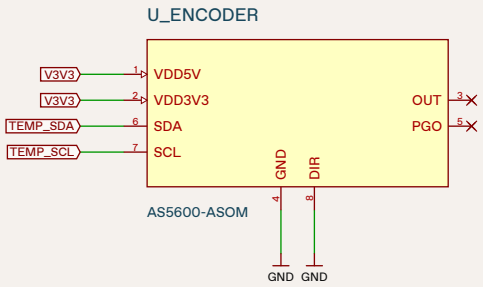


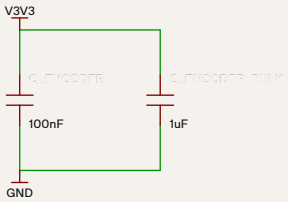
Optional encoder (DNP)

Bottom footprints only; AS5600 and both bypass capacitors omitted from factory assembly

Future noncontact angle feedback

DNP: no documented rear shaft; magnet and attachment undefined





U_ENCODER, C_ENCODER and C_ENCODER_BULK: bottom / DO NOT PLACE.

3.3 V mode: VDD5V + VDD3V3 tied together; DIR grounded; OUT/PGO NC.

Shared existing temperature I²C bus: AS5600 0x36 / TMP102 0x48.

Existing 4.7 k□ top-side pull-ups retained. Encoder firmware is required.

A diametrically magnetized shaft magnet and motor fit must be validated.